GROUP 03: Effect of Image Scaling and Zoom Augmentation

WEEK 4

1. OBJECTIVE FOR WEEK 4: To consolidate and compare the baseline and scaling/zoom-augmented training results for YOLOv8n, YOLOv11n, and YOLOv26n. The comparison includes Precision, Recall, mAP@50, mAP@50–95, training time, and inference time to understand the effect of different augmentation intensity levels on vehicle detection performance.

2. WORK COMPLETED

- Collected baseline performance metrics from Week 1.
- Collected performance metrics for Low, Medium, and High scaling/zoom augmentation from Week 3.
- Verified that the baseline and augmented datasets contain the same image filenames and split counts.
- Checked the training configuration for epochs, image size, batch size, and model variants.
- Consolidated baseline and augmented results into a single comparison matrix.
- Compared the performance of each YOLO version across the four experimental conditions.
- Identified the augmentation intensity that achieved the best performance for each model.

3. DATASET AND EXPERIMENTAL CONFIGURATION

3.1 Dataset Details

Parameter	**Details**
Total images	3,000
Training images	2,100 (70%)
Validation images	450 (15%)
Test images	450 (15%)
Number of classes	6: car, threewheel, bus, truck, motorbike, van
Image size	640 × 640
Annotation format	YOLO format

3.2 Experimental Settings

Parameter	Configuration
Model versions	YOLOv8n, YOLOv11n, YOLOv26n
Epochs / batch size	30 epochs / batch size 16
Augmentation type	Scaling / Zoom
Low intensity	0.90–1.10
Medium intensity	0.75–1.25
High intensity	0.60–1.40
Built-in augmentations	Disabled
Evaluation metrics shown below	Week 3 validation results; the report's source table labels these as validation results

The validation and test sets were kept unaugmented; scaling/zoom was applied to the training set. Hardware and software details are not repeated here because the supplied reports do not provide enough information to independently verify them.

4. DATASET SPLIT VERIFICATION

Split	Baseline	Low	Medium	High	Filename match
Train	2,100	2,100	2,100	2,100	Yes
Validation	450	450	450	450	Yes
Test	450	450	450	450	Yes

All four versions had the expected split counts, and the filename comparison returned True for each augmentation intensity. This confirms consistency at the filename level. The Week 3 report states that validation and test images were kept unchanged while augmentation was applied to training images.

5. CONSOLIDATED EXPERIMENTAL RESULTS

The augmented-run values below are copied from the Week 3 validation results table. Inference time is reported in seconds for the full test set in that source table, although the section is labelled as validation results; the measurement split should be confirmed before final submission. Baseline timing values and some baseline metrics were not present in the supplied Week 2/Week 3 reports, so they are marked N/A rather than estimated.

YOLO version	Intensity	Precision	Recall	mAP@50	mAP@50–95	Train (min)	Inference
YOLOv8n	Baseline	0.9267	0.8851	0.9329	0.8337	N/A	N/A
YOLOv8n	Low	0.9455	0.8716	0.9295	0.8254	17.2	10.5 s
YOLOv8n	Medium	0.9398	0.8642	0.9275	0.8281	17.0	11.8 s
YOLOv8n	High	0.9355	0.8636	0.9206	0.8189	17.0	12.1 s
YOLOv11n	Baseline	0.9131	0.8735	0.9189	0.8038	N/A	N/A
YOLOv11n	Low	0.9187	0.8800	0.9217	0.8140	19.1	11.4 s
YOLOv11n	Medium	0.9388	0.8806	0.9241	0.8245	17.7	9.8 s
YOLOv11n	High	0.9555	0.8785	0.9138	0.8084	18.0	10.7 s
YOLOv26n	Baseline	0.9370	0.8469	0.9176	0.8283	N/A	N/A
YOLOv26n	Low	0.9717	0.9103	0.9593	0.8694	19.8	11.4 s
YOLOv26n	Medium	0.9596	0.8973	0.9532	0.8614	20.0	11.2 s
YOLOv26n	High	0.9476	0.8980	0.9402	0.8606	20.3	9.6 s

Baseline values were taken from the Week 1 report. YOLOv26n baseline mAP@50 is 0.9176 and mAP@50–95 is 0.8283. Training and inference times were not reported in the Week 1 comparison table, so they remain N/A rather than being estimated.

6. PERFORMANCE COMPARISON AND OBSERVATIONS

6.1 YOLOv8n

Among the augmented runs, Low intensity gave the highest precision (0.9455) and mAP@50 (0.9295), while Medium gave the best mAP@50–95 (0.8281). Medium and High had the shortest reported training time at 17.0 minutes, compared with 17.2 minutes for Low. Compared with the baseline, all three augmentation levels increased precision but reduced recall and both mAP scores. For this model, scaling/zoom did not improve overall mAP over the baseline, although Low intensity produced the strongest augmented result.

6.2 YOLOv11n

Medium intensity was the strongest overall balance for YOLOv11n: it achieved the best augmented mAP@50 (0.9241), mAP@50–95 (0.8245), and recall (0.8806), and had the fastest training time (17.7 minutes) and inference time (9.8 seconds) among its three augmented runs. High intensity produced the highest precision (0.9555), but its mAP@50 and mAP@50–95 were lower than the baseline. Low and Medium improved both mAP metrics over the baseline, with Medium giving the clearest overall improvement.

6.3 YOLOv26n

Low intensity performed best on all four detection metrics among the YOLOv26n augmented runs: precision 0.9717, recall 0.9103, mAP@50 0.9593, and mAP@50–95 0.8694. Medium was slightly lower on all four metrics, while High had the quickest inference time at 9.6 seconds but lower precision and mAP. The available baseline data includes only precision and recall, so a complete baseline-to-augmentation comparison for mAP cannot be made from the supplied reports**.**

6.4 Overall Comparison

Across the augmented experiments, YOLOv26n with Low intensity was the strongest configuration, with the highest precision, recall, mAP@50, and mAP@50–95 of all nine augmented runs. YOLOv11n benefited most consistently from Medium intensity, which improved both mAP measures over its baseline. For YOLOv8n, all augmented runs had lower mAP values than baseline. For YOLOv26n, all three augmentation levels improved both mAP measures over baseline, although Low remained the best overall. High intensity gave YOLOv26n the shortest inference time (9.6 seconds), but not its best detection metrics.

7. ISSUES AND CHALLENGES

- Combining the results from nine augmented runs with the Week 1 baseline metrics required checking that the same metrics and model versions were being compared.
- The Week 3 table labels the results as validation results but describes inference time as seconds for the full test set, so the evaluation split used for timing should be confirmed.
- Some baseline fields, especially YOLOv26n mAP values and all baseline timing values, were not available in the supplied reports and should be added from the original Week 1 outputs.

8. PLAN FOR NEXT WEEK

- Generate precision–recall curves for all three YOLO versions.
- Generate confusion matrices and mAP comparison charts.
- Calculate class-wise Average Precision (AP).
- Plot training and validation loss curves for the runs.
- Collect sample detections showing correct predictions, false positives, and false negatives.
- Review why Low intensity worked best for YOLOv26n and why the results differed for YOLOv8n and YOLOv11n.
- Verify the missing baseline metrics and confirm the evaluation split used for inference-time measurements.

9. HOURS SPENT AND CONTRIBUTION PER MEMBER

Member	Hours Spent	Contribution
Simran Koshta (B230707CS)	3	Checked the baseline and augmented dataset details, verified split consistency, and contributed to consolidating the results.
Gunna Mohana Prathyusha (B230324CS)	4	Collected and organized the Low, Medium, and High augmentation results for the YOLO models and helped compare the performance metrics.
Maithili Kolasani (B230385CS)	3	Compared Precision, Recall, mAP@50, and mAP@50–95 across the different augmentation intensities and helped identify the best-performing configurations.
Sreeneha Bhavanasi (B230092CS)	4	Checked the training and inference time results, reviewed the model outputs, and helped compare the baseline and augmented models.
Ardhra Alaina Winston (B230198CS)	3	Consolidated the baseline and augmented metrics into a comparison table, summarized the observations for each YOLO version, and prepared the Week 4 report.